Lecture 21 — Systems, $Ax = b$, and elimination

Mon Oct 19

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Boyd video (optional): Lecture 25 — linear equations · Lecture 31 — solving linear equations · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 8.3, plus Gaussian elimination (class notes) and Lachniet Ch. 2 (solving systems). Related objectives: Lectures. Schedule. Connects to Ax as a linear combination of columns (Lecture 19).

One meeting for what had been two lectures. Board the warm-up, the fill-ins, and **W1**–**W4**. **W5**–**W7** are the same ideas if the room is fast. Hand work only; RREF and software are Wednesday.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, you should be able to:

1. Translate equations into $Ax = b$ and classify the system as square, overdetermined, or underdetermined.
2. Interpret $Ax = b$ as asking whether b is in the span of the columns, and anticipate no solution, one solution, or infinitely many.
3. Match each elementary row operation with its matrix, use those matrices to eliminate, and solve a small system by back substitution.
4. Explain why those row operations preserve the solution set.

Warm-up

For

$$\begin{aligned} x_1 - 2x_2 &= 3 \\ 2x_1 + x_2 &= 1, \end{aligned}$$

write A, x, b, and the augmented matrix $[A \mid b]$. Is b in the span of the columns of A?

Fill in

A linear system with m equations in n unknowns can be written $Ax = b$, where A is _____, x is an _____-vector, and b is an _____-vector.

Column view. Solving $Ax = b$ asks: is b a _____ of the columns of A? Equivalently: does b lie in span{columns of A}?

Shape vocabulary (compare m equations to n unknowns):

Name	Relation	Typical intuition
Square	$m =$ ____	Often one solution, but not always
Overdetermined	$m >$ ____	Often _____ (too many constraints)
Underdetermined	$m <$ ____	Often _____ (free variables)

These are **tendencies**, not guarantees. A square system can be inconsistent or have infinitely many solutions. An overdetermined system can still be consistent.

Possible outcomes for $Ax = b$:

1. **No solution** — b is _____ the column space.
2. **Unique solution** — b is in the column space in exactly _____ way.
3. **Infinitely many solutions** — b is in the column space in _____ ways (free parameters).

The **augmented matrix** of $Ax = b$ is $[A \mid b]$, with b as an extra _____.

Elementary row operations that do not change the solution set:

1. _____ two rows.
2. Multiply a row by a _____ scalar. (Multiplying by 0 is not allowed: it is not reversible and can destroy an equation.)
3. Add a _____ of one row to another row.

Matrices for those operations. Left-multiplying $[A \mid b]$ by a matrix does the row operation. Build that matrix by doing the same operation to I. For two equations:

Swap the two rows:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Multiply row 1 by $c \neq 0$, or multiply row 2 by $c \neq 0$:

$$\begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix}.$$

Add c times row 1 to row 2, or add c times row 2 to row 1:

$$\begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}, \qquad \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}.$$

The extra number sits in the row you are changing, in the column of the row you copy from. For three equations, start from I_3 and edit that same spot. Adding 2 times row 1 to row 3 uses

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}.$$

Why they preserve solutions: each allowed operation is reversible, so the new system is _____ to the old one. Each matrix above is invertible, which is the same fact in matrix language.

Gaussian elimination aims for **echelon form**: zeros _____ each pivot, pivots marching right as you go down. Then **back substitution** solves from the _____ equation upward. Reduced form, free variables, and software are next time.

Worked in class

W1. Matrix form. Three unknowns x_1, x_2, x_3:

$$\begin{aligned} x_1 + x_2 &= 4 \\ x_2 - x_3 &= 1 \\ 2x_1 + x_3 &= 3. \end{aligned}$$

Write A, x, and b. Square / over / under?

W2. Span interpretation. Let

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \quad c = \begin{bmatrix} 3 \\ 5 \end{bmatrix}.$$

(a) Are the columns of A independent? Describe span{columns}. (b) Does $Ax = b$ have a solution? Infinitely many? (c) Does $Ax = c$ have a solution?

W3. Eliminate and back-substitute. Starting from

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & 3 & 8 \end{array}\right],$$

replace Row 2 by (Row 2) $-$ 2·(Row 1). Solve for x_2, then x_1.

W4. Three operations, then a full solve by applying matrices. Start from the warm-up matrix

$$M = \left[\begin{array}{cc|c} 1 & -2 & 3 \\ 2 & 1 & 1 \end{array}\right].$$

Swap the two rows:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} M = \left[\begin{array}{cc|c} 2 & 1 & 1 \\ 1 & -2 & 3 \end{array}\right].$$

Multiply row 1 by 2:

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} M = \left[\begin{array}{cc|c} 2 & -4 & 6 \\ 2 & 1 & 1 \end{array}\right].$$

Add -2 times row 1 to row 2. This is the elimination step:

$$E_1 = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}, \qquad E_1 M = \left[\begin{array}{cc|c} 1 & -2 & 3 \\ 0 & 5 & -5 \end{array}\right].$$

Continue from $E_1 M$ until the left side is I. Multiply the new row 2 by $\frac{1}{5}$:

$$E_2 = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{5} \end{bmatrix}, \qquad E_2 E_1 M = \left[\begin{array}{cc|c} 1 & -2 & 3 \\ 0 & 1 & -1 \end{array}\right].$$

Add 2 times the new row 2 to row 1:

$$E_3 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \qquad E_3 E_2 E_1 M = \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -1 \end{array}\right].$$

Read the solution $x = (1, -1)$. The same three matrices clear A itself: $E_3 E_2 E_1 A = I$. We name that product A^{-1} later.

W5 (if time). Anticipating outcomes. Without fully solving, say what you expect (and why):

(a) 2 equations, 2 unknowns, with clearly parallel incompatible lines.

(b) 1 equation, 3 unknowns: $x_1 + x_2 + x_3 = 5$.

(c) 3 equations, 2 unknowns that happen to be consistent (the third is the sum of the first two).

W6 (if time). Context $\rightarrow Ax = b$. Two foods (oats, yogurt) and two nutrients (protein, fiber). Per serving: oats give 2 g protein and 1 g fiber; yogurt gives 3 g protein and 1 g fiber. Monday's targets: 26 g protein and 10 g fiber. Write the two equations and the matrix form $Ax = b$. Square? (Preview of Lab 3.)

W7 (if time). A 3×3 echelon. Reduce

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 5 \\ 2 & 1 & 0 & 5 \end{array}\right]$$

to echelon form and decide whether a solution exists.

Your turn

Start with Y1–Y4. Y5–Y9 are the elimination cousins if you finish.

Y1. Write as $Ax = b$:

$$\begin{aligned} 3x_1 - x_2 &= 0 \\ x_1 + 4x_2 &= 7. \end{aligned}$$

Classify as square / over / under.

Y2. Let A have columns $a_1 = (1, 0, 1)$ and $a_2 = (0, 1, 1)$. For $b = (1, 1, 2)$, explain in one sentence why $Ax = b$ has a solution, and give one solution x.

Y3. True or false, with a one-line reason: every square system $Ax = b$ has exactly one solution.

Y4. A sensor gives 4 linear measurements for 2 unknown parameters. Is the system typically overdetermined? Does that guarantee inconsistency?

Y5. Augmented matrix for

$$x_1 - x_2 = 1$$
$$3x_1 + 2x_2 = 12.$$

Y6. Clear the $(2,1)$ entry of

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 3 & 2 & 12 \end{array}\right]$$

by a row replacement. Write the new matrix.

Y7. From the echelon form

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 5 & 9 \end{array}\right],$$

solve by back substitution.

Y8. True or false, with a one-line reason: if you swap two rows of $[A \mid b]$, the solution set of $Ax = b$ can change.

Y9. Write the 2×2 elementary matrix for each operation.

(a) Multiply row 1 by 4.

(b) Add 3 times row 2 to row 1.

(c) Apply the matrix from (b) to $\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 5 \end{array}\right]$.

Answers (Your turn)

Y1.

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 4 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 7 \end{bmatrix}.$$

Square ($m = n = 2$).

Y2. $b = 1 \cdot a_1 + 1 \cdot a_2$, so b is in the column span; one solution is $x = (1, 1)$.

Y3. **False.** Square systems can be inconsistent (parallel lines) or have infinitely many solutions (the same line twice).

Y4. Typically overdetermined ($m = 4 > n = 2$). No — consistency depends on whether b lands in the column space.

Y5.

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 3 & 2 & 12 \end{array}\right].$$

Y6. Row2 $\leftarrow$ Row2 $-$ 3·Row1 gives

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 5 & 9 \end{array}\right].$$

Y7. From Row 2: $5x_2 = 9$ so $x_2 = 9/5$. From Row 1: $x_1 - x_2 = 1$ so $x_1 = 1 + 9/5 = 14/5$. Solution $x = (14/5,\ 9/5)$.

Y8. **False.** Swapping rows reorders equations and keeps the same solution set.

Y9. (a) $\begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$. (b) $\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$. (c) Row 1 becomes $(1, 0, 2) + 3(0, 1, 5) = (1, 3, 17)$, so

$$\left[\begin{array}{cc|c} 1 & 3 & 17 \\ 0 & 1 & 5 \end{array}\right].$$